

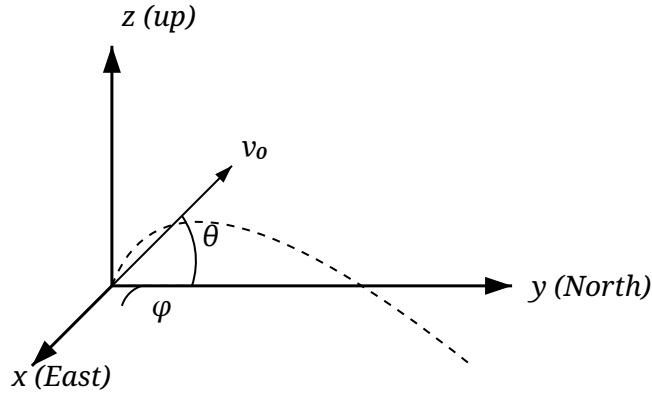


Figure 15.6 A projectile fired East of North. Note that ϕ is measured from North. The elevation angle θ is measured from the horizontal plane.

(b) The Coriolis acceleration is

$$\begin{aligned}\mathbf{a}_c &= -2\boldsymbol{\Omega} \times \mathbf{v} = -2 \left[\Omega \sin \lambda \hat{\mathbf{j}} + \Omega \cos \lambda \hat{\mathbf{k}} \right] \times \left[v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}} + v_z \hat{\mathbf{k}} \right], \\ &= -2 \left[-v_x \Omega \sin \lambda \hat{\mathbf{k}} + v_z \Omega \sin \lambda \hat{\mathbf{i}} + v_x \Omega \cos \lambda \hat{\mathbf{j}} - v_y \Omega \cos \lambda \hat{\mathbf{i}} \right].\end{aligned}$$

Since $v_x = v_{0x}$, $v_y = v_{0y}$ and $v_z = v_{0z} - gt$ we obtain

$$\begin{aligned}a_x &= 2v_0 \Omega \cos \lambda \cos \theta \cos \phi - 2\Omega \sin \lambda (v_0 \sin \theta - gt), \\ a_y &= -2v_0 \Omega \cos \lambda \cos \theta \sin \phi, \\ a_z &= 2v_0 \Omega \sin \lambda \cos \theta \sin \phi - g.\end{aligned}$$

(c) The velocity is determined by integrating the acceleration components to obtain:

$$\begin{aligned}v_x &= v_{0x} + \int a_x dt \\ &= v_0 \cos \theta \sin \phi + (2v_0 \Omega \cos \lambda \cos \theta \cos \phi)t \\ &\quad - (2v_0 \Omega \sin \lambda \sin \theta)t + gt^2 \Omega \sin \lambda, \\ v_y &= v_{0y} + \int a_y dt = v_0 \cos \theta \cos \phi - (2v_0 \Omega \cos \lambda \cos \theta \sin \phi)t, \\ v_z &= v_{0z} + \int a_z dt = v_0 \sin \theta - gt + (2v_0 \Omega \sin \lambda \cos \theta \sin \phi)t.\end{aligned}$$

Exercise 15.6

Carry out the operation $\boldsymbol{\Omega} \times \mathbf{v}$ to verify Equation (15.7).

Exercise 15.7

A projectile located at 60°N is fired due East with an initial velocity of 300 m/s at an angle 25° above the horizontal. Evaluate the initial values of the components of the Coriolis acceleration. (Note: $\lambda = 30^\circ$.) Answer:
 $a_x(t = 0) = -9.22 \times 10^{-3}\text{ m/s}^2$.

15.6 The Foucault Pendulum

The Foucault pendulum is an important example of the effect of a rotating coordinate system on the dynamics of a physical system. This pendulum consists of a massive bob suspended by a very

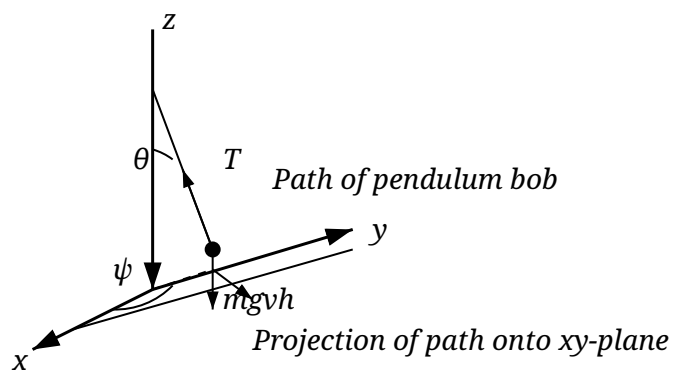


Figure 15.7 The Foucault pendulum.

long string or thin wire. Foucault pendulums range from ten to sixty meters in length. When set in motion the bob swings back and forth and the plane of motion of the pendulum slowly precesses. The precession rate is $\Omega \cos \lambda$, where Ω is the angular velocity of the Earth ($2\pi\text{ rad/day}$) and λ is the colatitude. A long pendulum goes through many oscillations before the motion is damped out, and the precession

is quite noticeable. When Foucault suspended a pendulum inside the Pantheon in Paris large crowds showed up to "observe the rotation of the Earth."

The behavior of the Foucault pendulum is easy to understand if you imagine a pendulum suspended above the North Pole. The Earth rotates under the pendulum and an observer standing on the surface sees the plane of motion precessing at a rate of one revolution per day. On the other hand, at the equator, a pendulum does not precess at all. The motion at an arbitrary latitude is not intuitively obvious and requires an analysis to understand it.

Consider a long pendulum at an arbitrary latitude. Let the z -axis be oriented along the local vertical. See [Figure 15.7](#), which shows the projection of the path of the pendulum onto the xy -plane. The projected path makes an angle ψ with the x -axis. The precession of the pendulum causes ψ to change as a function of time. The basic problem consists in demonstrating that the rate of change of ψ is

$$\frac{d\psi}{dt} = \Omega \cos \lambda.$$

The pendulum bob is acted upon by two real forces: the tension in the string ($\mathbf{T}$) and gravity ($m\mathbf{g}$). It is also acted upon by two fictitious forces: the Coriolis force ($-2m\boldsymbol{\Omega} \times \mathbf{v}$) and the centrifugal force. For convenience assume the centrifugal force is absorbed into $m\mathbf{g}$. Therefore,

$$m\mathbf{a} = \mathbf{T} + m\mathbf{g} - 2m(\boldsymbol{\Omega} \times \mathbf{v}).$$

In component form this relationship becomes

$$\begin{aligned} ma_x \hat{\mathbf{i}} + ma_y \hat{\mathbf{j}} + ma_z \hat{\mathbf{k}} = & -T \sin \theta \cos \psi \hat{\mathbf{i}} - T \sin \theta \sin \psi \hat{\mathbf{j}} + T \cos \theta \hat{\mathbf{k}} - mg \hat{\mathbf{k}} \\ & - 2m \left[\Omega \cos \lambda \hat{\mathbf{k}} + \Omega \sin \lambda \hat{\mathbf{j}} \right] \times \left[v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}} + v_z \hat{\mathbf{k}} \right]. \end{aligned}$$

That is,

$$-T \sin \theta \cos \psi + 2mv_y \Omega \cos \lambda - 2mv_z \Omega \sin \lambda = ma_x, \quad (15.8)$$

$$-T \sin \theta \sin \psi - 2mv_x \Omega \cos \lambda = ma_y, \quad (15.9)$$

$$T \cos \theta - mg + 2mv_x \Omega \sin \lambda = ma_z. \quad (15.10)$$

Consider Equations (15.8) and (15.9). Because the vertical displacement is small,

the vertical velocity v_z is much smaller than the horizontal velocity. Therefore, we are justified in ignoring the term containing v_z in Equation (15.8), leading to the set of coupled differential equations

$$\begin{aligned} -T \sin \theta \cos \psi + 2mv_y \Omega \cos \lambda &= ma_x, \\ -T \sin \theta \sin \psi - 2mv_x \Omega \cos \lambda &= ma_y. \end{aligned}$$

The pendulum is very long so the angle θ is quite small and the string is essentially vertical at all times (Figure 15.7 is significantly out of scale). To a high degree of accuracy the tension in the string is equal to mg . Furthermore, $x = l \sin \theta \cos \psi$ and $y = l \sin \theta \sin \psi$. Therefore, the coupled equations above can be formulated as:

$$\begin{aligned} -\left(\frac{g}{l}\right)x + 2v_y \Omega \cos \lambda &= a_x, \\ -\left(\frac{g}{l}\right)y - 2v_x \Omega \cos \lambda &= a_y. \end{aligned}$$

Noting that $g/l = \omega^2$, where ω is the oscillatory frequency of the pendulum, and letting $K = \Omega \cos \lambda$, these equations become

$$\begin{aligned} \ddot{x} - 2K\dot{y} + \omega^2 x &= 0, \quad (15.11) \\ \ddot{y} + 2K\dot{x} + \omega^2 y &= 0. \end{aligned}$$

The problem now is to solve these coupled linear differential equations. I will do this in three ways. First I will use a "trick" then I will solve the equations in a straightforward, but more tedious manner. Then, in Worked Example 15.5, I will show you a third way to find the solution. The reason for doing this is because I want you to see several techniques for solving coupled differential equations.

A Neat Trick

The set of coupled equations (15.11) can be solved very nicely by introducing the complex quantity

$$\zeta = x + iy.$$

Multiplying the second equation by $i(= \sqrt{-1})$ and adding the two equations gives

$$\ddot{\zeta} + 2iK\dot{\zeta} + \omega^2 \zeta = 0,$$

which resembles the equation of motion for a damped harmonic oscillator. This linear differential equation is solved in the usual way, assuming $\zeta = e^{pt}$ and obtaining the "secular" or "characteristic" equation

$$p^2 + 2iKp + \omega^2 = 0,$$

with solution

$$p = -iK \pm i\sqrt{K^2 + \omega^2}.$$

Now $K = \Omega \cos \lambda$, where Ω is the rotation rate of the Earth, whereas $\omega = \sqrt{g/l}$ is the angular frequency of the pendulum. So $K \ll \omega$ and the K^2 under the root can be neglected. There are two possible values for p , so the solution is

$$\zeta = Ae^{-i(K+\omega)t} + Be^{-i(K-\omega)t}, \quad (15.12)$$

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where A and B are complex constants. Writing the complex constants in the form $(a + ib)$ and $(c + id)$ the solution is

$$\zeta(t) = (a + ib)e^{-i(K+\omega)t} + (c + id)e^{-i(K-\omega)t}. \quad (15.13)$$

This complex solution is expressed in terms of real quantities, but it is still not obvious what the motion looks like. The following example will help you to visualize the motion described by [Equation \(15.13\)](#).

Worked Example 15.4 At time $t = 0$ the Foucault pendulum is released from rest at position $x = \mathcal{A}$, $y = 0$. Obtain expressions for $x(t)$ and $y(t)$.

Solution Let us write the constants in [Equation \(15.12\)](#) as $\frac{1}{2}Ce^{\pm i\theta}$, thus

$$\begin{aligned} \zeta &= \frac{1}{2}Ce^{-i\theta}e^{-i(\omega+K)t} + \frac{1}{2}Ce^{i\theta}e^{i(\omega-K)t} \\ &= \frac{1}{2}Ce^{-iKt} \left[e^{-i(\omega t + \theta)} + e^{i(\omega t + \theta)} \right] \\ &= \frac{1}{2}C[\cos Kt - i \sin Kt][2 \cos(\omega t + \theta)] \\ &= C[\cos Kt \cos(\omega t + \theta) - i \sin Kt \cos(\omega t + \theta)]. \end{aligned}$$

Because $x = \text{Re}(\zeta)$ and $y = \text{Im}(\zeta)$ we conclude that

$$\begin{aligned} x &= C \cos(t\Omega \cos \lambda) \cos(\omega t + \theta), \\ y &= -C \sin(t\Omega \cos \lambda) \cos(\omega t + \theta). \end{aligned}$$

The constants C and θ are now easily obtained from the initial conditions, yielding $C = \mathcal{A}$ and $\theta = 0$, so

$$x(t) = [\mathcal{A} \cos(t\Omega \cos \lambda)] \cos \omega t,$$

$$y(t) = -[\mathcal{A} \sin(t\Omega \cos \lambda)] \cos \omega t.$$

The motion is a rapid oscillation of frequency ω , modulated by a low-frequency variation in amplitude ($\Omega \cos \lambda$). The modulated x and y oscillations have time varying amplitudes that are 180° out of phase. The resultant motion is a rotation of the plane of motion with a frequency $\Omega \cos \lambda$. This is illustrated in Figure 15.8.

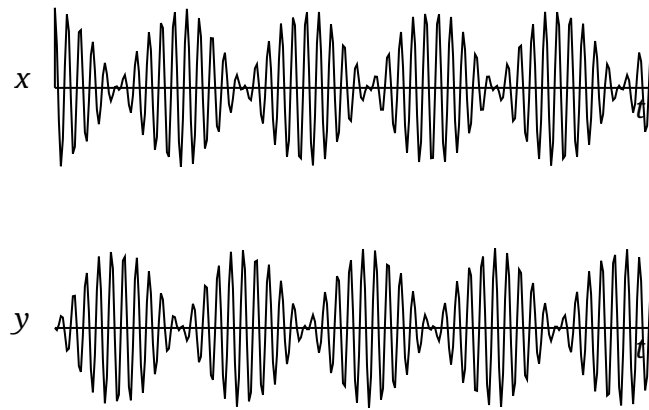


Figure 15.8 Amplitudes of oscillations in the x and y directions for the Foucault pendulum.

Exercise 15.8

A Foucault pendulum is observed to have a velocity v_0 in the $+x$ -direction at time $t = 0$ when it passes through the origin. Obtain expressions for x and y as functions of t . Answer: $x = \text{Re}(\zeta) = \frac{v_0}{\omega} \sin \omega t \cos Kt$,

$$y = \text{Im}(\zeta) = \frac{v_0}{\omega} \sin \omega t \sin Kt.$$

A More General Way to Solve the Problem

Return again to [Equations \(15.11\)](#). They are a pair of coupled second-order linear differential equations whose solution is desired. By this time you are very familiar with the technique of assuming a solution of the form Ae^{pt} . In this case there are two unknowns and the procedure is to assume two solutions, but to let both of them have the same time dependence.⁵ In other words, assume the solutions

$$\begin{aligned}x &= Ae^{pt}, \\y &= Be^{pt}.\end{aligned}$$

Plugging these solutions into [Equations \(15.11\)](#) leads to the following set of secular equations

$$\begin{aligned}p^2 A - 2KpB + \omega^2 A &= 0, \\p^2 B + 2KpA + \omega^2 B &= 0.\end{aligned}$$

or

$$\begin{aligned}A(p^2 + \omega^2) - B(2Kp) &= 0, \\A(2Kp) + B(p^2 + \omega^2) &= 0.\end{aligned}$$

This is a system of homogeneous linear algebraic equations in the unknowns A and B . Recall that the condition for a nontrivial solution is

$$\begin{vmatrix} p^2 + \omega^2 & -2Kp \\ 2Kp & p^2 + \omega^2 \end{vmatrix} = 0.$$

This leads to

$$p = -iK \pm i\sqrt{K^2 + \omega^2}. \quad (15.14)$$

The rest of the solution is straightforward.⁶

Worked Example 15.5 Solve for the precession rate of a Foucault pendulum by considering the problem in a rotating reference frame in which the pendulum does not precess.

Solution Using a rotating reference frame makes the problem easier to solve. Imagine looking down on the pendulum from a coordinate system rotating at an angular velocity $K = \Omega \cos \lambda$, equal to the precession rate. In this coordinate frame the bob moves back and forth in a straight line. Let the axes in this coordinate frame be x^* and y^* and define the x^* -axis to lie along the path of the bob. Then y^* is always zero. Denoting the distance from the origin to the bob along the x^* -axis by s , [Figure 15.9](#) shows that the Earth-based

coordinates x and y are

$$x = s \cos Kt$$

⁵ You probably remember that this is the procedure we used to solve the coupled oscillators problem. (See [Section 11.5](#).)

⁶ It is interesting to note that the Foucault pendulum problem has many parallels to a seemingly completely unrelated problem, namely the violation of CP symmetry in the decay of an elementary particle called the neutral kaon. To explore further, see the paper "Classical Illustration of CP Violation in Kaon Decays" by Jonathan L. Rosner and Scott A. Slezak, *American Journal of Physics*, **69**, 44–49 (2001).

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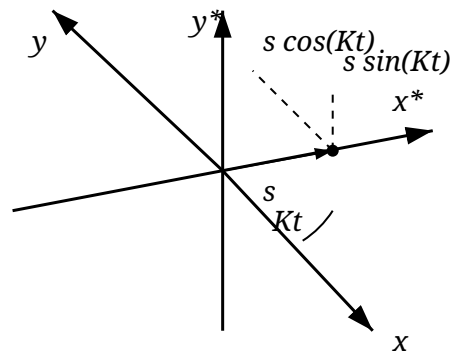


Figure 15.9 The starred coordinate system rotates at an angular velocity $K = \Omega \cos \lambda$ relative to the unstarred system. The motion of the pendulum bob is along the x^* -axis.

and

$$y = -s \sin Kt.$$

The sum of the equations of motion, [Equations \(15.11\)](#), is

$$\ddot{x} + \ddot{y} + 2K(\dot{x} - \dot{y}) + \omega^2(x + y) = 0 \quad (15.15)$$

Plugging in the expressions for x and y in terms of s gives

$$\ddot{s} + (K^2 + \omega^2)s = 0. \quad (15.16)$$

Again, $K \ll \omega$ so, to an excellent approximation,

$$\ddot{s} + \omega^2 s = 0.$$

This is just the SHM equation. Thus, in the rotating coordinate system the Foucault pendulum simply oscillates back and forth, or, from the point of view of the observer on the surface of the Earth, the plane of the pendulum rotates at a constant rate $K = \Omega \cos \lambda$.

Exercise 15.9

Fill in the missing steps between [Equations \(15.15\) and \(15.16\)](#).

Exercise 15.10

Show that [Equation \(15.14\)](#) leads to the same result as obtained in Worked Example 15.5.

Exercise 15.11

Suppose that a 20 m Foucault pendulum is mounted at San Francisco, California (latitude 37.77°). Determine the precession rate of its plane of motion. Answer: $11.9^\circ/\text{h}$.

15.7 Application: The Tidal Force (Optional)

The tidal force is an interesting and important astronomical phenomenon. Tides are responsible for the formation of rings around large planets, tidal locking (or resonances) between bodies, the stretching of bodies falling into black holes ("spaghettization"), the frictional heating of Jupiter's

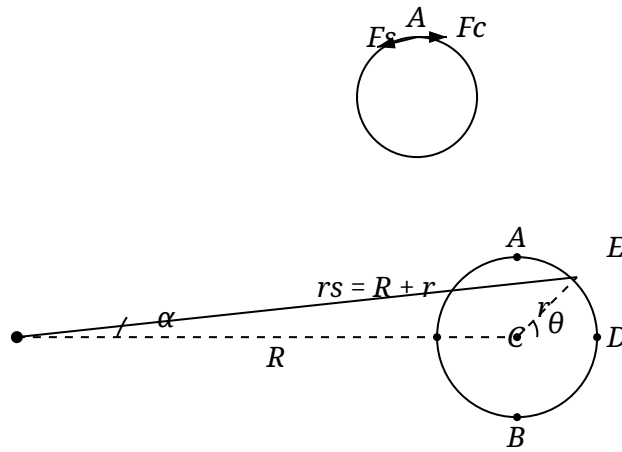


Figure 15.10 The lower figure gives the geometrical parameters. The upper figure shows that forces at point A do not cancel.

moon Io, and moonquakes, to mention a few tidal effects. In [Chapter 7](#) we noted that the ocean tides on the Earth, due primarily to the Moon, are slowing down the rotation rate of the Earth.

A particularly interesting aspect of tidal forces is that they are due to the inhomogeneity of the gravitational field acting on a body and consequently are given by an inverse cube force law.

To analyze the tidal force, we shall consider the tides raised by the Sun on the Earth's oceans. Although the lunar tides are nearly twice as big as the solar tides, the solar tides are easier to analyze (because the center of mass of the system is essentially at the center of the Sun) and the physical principles are the same. The equations we will derive are valid for both situations, although the numerical values will be different.

We begin by assuming a solid rigid Earth that is not rotating about its axis, but it is revolving about the Sun in a circle at angular velocity Ω . Thus a coordinate system fixed in the Earth has its axes pointing towards the same fixed stars as it orbits the Sun. This has been described as the same motion a cook uses in moving a frying pan in a circle while holding the handle. Of course, the Earth really is rotating and we will include that fact a little later, but the picture of the nonrotating Earth allows us to clearly distinguish the two tidal bulges. The geometry of the system is shown in the bottom part of [Figure 15.10](#).

In this coordinate system the forces acting on a particle *at the center of the Earth* are the attractive force of the Sun ($\mathbf{F}_S = -G \frac{Mm}{R^2} \hat{\mathbf{R}}$) and the (fictitious) centrifugal force ($\mathbf{F}_c = -m\Omega \times (\Omega \times \mathbf{R})$), where $\mathbf{R}$ is the vector from Sun to Earth, M is the mass of the Sun and m is the mass of the particle. These forces are equal and

opposite.

The Earth is assumed to be perfectly rigid so all points in the Earth are accelerating at the same rate. That is, every part of the Earth feels the same centrifugal force. But not all points feel the same gravitational attraction toward the Sun. As indicated in the top part of Figure 15.10, at point A the two forces are not perfectly antiparallel and there is a small unbalanced component directed toward the center of the globe. Such forces, due to the difference between $\mathbf{F}_s$ and $\mathbf{F}_c$ are called "tidal forces." Similarly, at point B there is a net force toward the center. Point C is nearer to the Sun than the center of the planet, so the solar attraction is greater than the centrifugal force, resulting in a small unbalanced (tidal) component directed away from the center. At point D the centrifugal force is greater than the solar force so once again the tidal force is away from the center. In other words, at points A and B there is a compression and at points C and D the tidal force exerts a stretching force on the Earth. (It might be helpful to consider the behavior of a spherical gas cloud falling towards the Sun. The same forces are acting on the gas cloud, but it is

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not rigid so the cloud is squeezed and stretched into a long thin shape. This would be an example of spaghettization.)

Let us now determine the tidal force at an arbitrary point on Earth's surface such as point E in the bottom part of Figure 15.10. Because the centrifugal force at the center of the Earth is equal and opposite to the Sun's gravitational attraction we can write

$$\mathbf{F}_c = \frac{GMm}{R^3} \mathbf{R}.$$

At point E the gravitational force depends on the distance to the Sun, $\mathbf{r}_s$.

$$\mathbf{F}_s = -\frac{GMm}{r_s^3} \mathbf{r}_s.$$

Therefore, the tidal force, being the difference between these, is

$$\mathbf{F}_{Tide} = \mathbf{F}_c - \mathbf{F}_s = -GMm \left(\frac{\mathbf{r}_s}{r_s^3} - \frac{\mathbf{R}}{R^3} \right).$$

Let us now determine r_s^{-3} .

Note that $r_s^2 = (\mathbf{R} + \mathbf{r}) \cdot (\mathbf{R} + \mathbf{r}) = R^2 + r^2 + 2(\mathbf{R} \cdot \mathbf{r})$. Since $R^2 \gg r^2$ we can

write

$$\begin{aligned}
 r_s^2 &\approx R^2 \left(1 + 2 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right) \\
 \therefore 1/r_s^3 &= (r_s^2)^{-3/2} = \left[R^2 \left(1 + 2 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right) \right]^{-3/2} \\
 &\doteq \frac{1}{R^3} \left(1 - 3 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right),
 \end{aligned}$$

so

$$\begin{aligned}
 \mathbf{F}_{Tide} &= -GMm \left[\frac{\mathbf{r}_s}{R^3} \left(1 - 3 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right) - \frac{\mathbf{R}}{R^3} \right] \\
 &= -\frac{GMm}{R^3} \left[(\mathbf{R} + \mathbf{r}) \left(1 - 3 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right) - \mathbf{R} \right] \\
 &= -\frac{GMm}{R^3} \left(\mathbf{R} + \mathbf{r} - 3 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} (\mathbf{R} + \mathbf{r}) - \mathbf{R} \right).
 \end{aligned}$$

But $\mathbf{R} + \mathbf{r} \approx \mathbf{R}$, so

$$\mathbf{F}_{Tide} = -\frac{GMm}{R^3} \left(\mathbf{r} - 3\mathbf{R} \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right).$$

Observe that the main contributions of $\mathbf{F}_s$ and $\mathbf{F}_c$ have cancelled, leaving only their difference. At points A and B we have $\mathbf{R} \perp \mathbf{r}$ so $\mathbf{F}_{Tide} = -\frac{GMm}{R^3} \mathbf{r}$ which is pointing towards the center of the Earth, i.e., a compression. At point C the angle θ is π so and we obtain $\mathbf{F}_{Tide} = +2\frac{GMm}{R^3} \mathbf{r}$ which points away from the surface. A similar expression is obtained at D.

The tidal forces at various points on the surface of the globe are shown in [Figure 15.11](#).

At a general point, such as E in [Figure 15.11](#), the tidal force can be resolved into components perpendicular to the surface (along $\hat{\mathbf{r}}$) and parallel to the surface (along $\hat{\theta}$). The components perpendicular to the surface (such as at ABCD) simply add to the Earth's gravitational force and are not of interest. But the tangential components can affect particles on the Earth's surface that are not rigidly attached to it and these are the forces that produce the tides in the ocean and atmosphere.